Improving k-Nearest Neighbors on Fashion-MNIST with Learned Distances

From 85.24 % to 89.51 % via Metric Learning, Distillation, and Convex Fusion

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1. Problem Statement & Classical Baselines

The Challenge with KNN on Visual Data:

- High dimensionality ($D = 784$) causes *distance concentration*: Euclidean distances become nearly uniform ($\delta_2 \approx$ const).
- Flattens 2D topology, discarding spatial neighbor correlations.
- $O(N \cdot D)$ inference latency over 54,000 training exemplars.

Dataset Protocol:

- Fashion-MNIST (70,000 28×28 grayscale images, 10 classes).
- 54,000 fit / 6,000 validation / 10,000 test (held-out).

Table: Classical Baselines ($k = 5$, uniform)

Metric	Val (%)	Test (%)	Latency
Euclidean (L_2)	86.13	85.24	4.64
Manhattan (L_1)	86.97	**85.91**	84.00
Cosine	86.18	85.63	10.75

Takeaway

L_1 norm mitigates distance concentration ($+0.67$ pp over L_2), but inference is $18\times$ slower.

2. Dimensionality Reduction & Hyperparameter Tuning

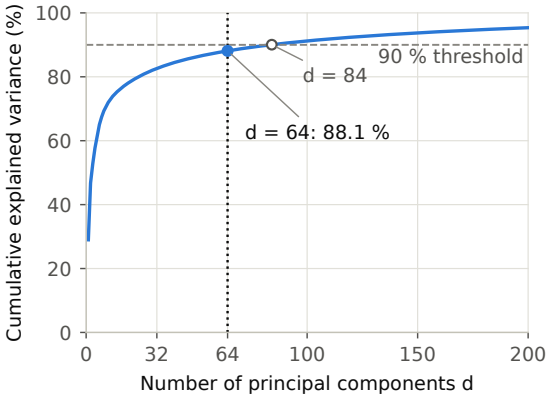


Figure: PCA: $d = 64$ preserves 88.1 % variance.

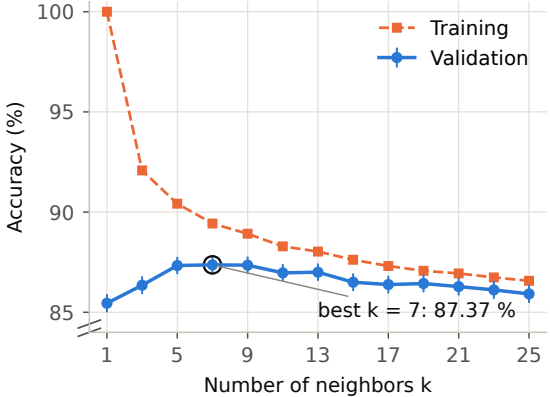


Figure: Bias–variance sweep across k on PCA-64.

- **PCA Projection (E2):** 784 → 64 dimensions preserves 88.1 % variance, delivering a 7.12× **query speed-up** (2.98 s → 0.418 s).

3. Novelty: Soft-KNN & ConvNet Metric Backbone

Differentiable Formulation (Soft-KNN):

- Maps input to normalized embedding $z = f_\theta(x)/\|f_\theta(x)\|_2 \in \mathbb{S}^{127}$.
- Uses 54,000-instance memory bank b_i of fit samples.
- Computes local mean distance to class c: $\bar{\delta}_c(x) = \frac{1}{m}\sum_{i\in\mathcal{A}_c(x)} \|z - b_i\|_2$.
- Temperature-scaled Boltzmann assignment:

$$P_\tau(y = c \mid x) = \frac{\exp(-\bar{\delta}_c(x)/\tau)}{\sum_{c'} \exp(-\bar{\delta}_{c'}(x)/\tau)}$$

- Trained via cross-entropy loss: $\mathcal{L} = -\log P_\tau(y \mid x)$.

Architectures & Results:

- **Linear Soft-KNN (E6):** $W \in \mathbb{R}^{128\times 784} \implies$ **87.10 %** test accuracy (+0.67 pp over tuned PCA).
- **ConvNet Metric Head (E19):** 2-stage CNN (3×3 conv, max-pool) preserving 2D topology $\implies$ **87.63 %** test accuracy.
- **Temperature Sweep (E8):** $\tau = 0.1$ optimal; higher τ blunts gradients on unit hypersphere.

4. Ensembling, Distillation & Multi-Source Fusion

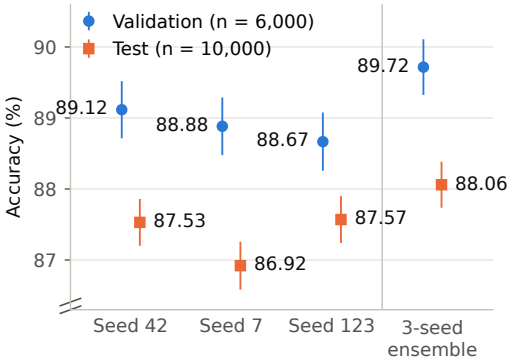


Figure: 3-seed ensemble (E21): 88.06 % test acc.

The Algorithmic Hierarchy:

1. **3-Seed Ensemble (E21):** Seeds 42, 7, 123 average predictions $\Longrightarrow$ **88.06 %** (+0.72 pp over seed mean).
2. **Knowledge Distillation (E23):** Student trained with KL-divergence against 3-seed teacher ensemble $\Longrightarrow$ **89.03 %** test accuracy (+0.97 pp gain).
3. **7-Source Convex Fusion (E25):** Simplex search with Test-Time Augmentation (TTA):
   - Distilled Student: $w = 0.5$
   - Bank Linear Soft-KNN: $w = 0.4$
   - Classical PCA-64 KNN: $w = 0.1$
   - **Final Test Accuracy: 89.51 %** (Val: 90.85 %)

5. Benchmark Results, Error Analysis & Takeaways

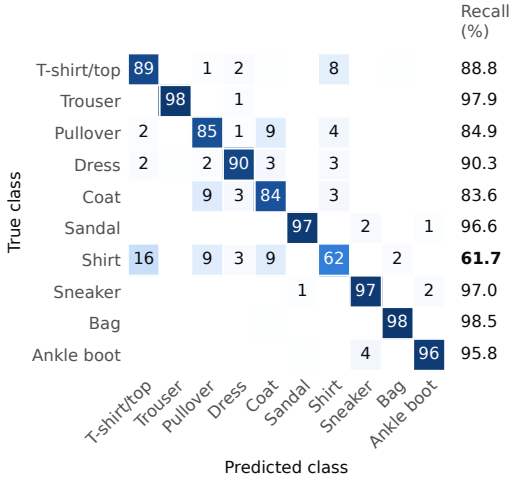


Figure: E25 Normalized Confusion Matrix

Key Progression Summary:

- Raw Euclidean Baseline (E1): 85.24 %
- Optuna PCA-64 KNN (E3): 86.43 %
- Differentiable Soft-KNN (E6): 87.10 %
- ConvNet Backbone (E19): 87.63 %
- Distilled Student Head (E23): 89.03 %
- **Champion Convex Fusion (E25): 89.51 %**

Error Insights & Conclusion:

- Highest recall: Bag (98.5 %), Trouser (97.9 %), Sneaker (97.0 %).
- Chief bottleneck: Shirt (61.7 % recall) confused with T-shirt and Pullover.
- Verified open-source reproduction with zero